

SC1001: Introduction to Computational Thinking

Tutorial 1: Flowcharts, Pseudocode, Data Types, and Variables

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Me

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- Formerly at the University of Maryland and the University of Colorado
- Research: natural language processing, question answering, topic models, and human–AI interaction
- Born in Colorado; grew up in Iowa
- High school in Arkansas
- Undergraduate study in California
- Graduate school in New Jersey
- Worked on an electronic dictionary in Berlin and Google Books in New York
- You can call me Jordan, JBG, or Boyd-Graber

What We Are Practising Today

- Trace and repair an algorithm
- If statements
- FizzBuzz
- Distinguish Python expressions, assignments, and errors
- Design an algorithm before writing Python
- Use a simple heuristic when an exact solution is unnecessary

Discussion 1: What Is Wrong?

The algorithm should read 10 scores and compute their average.

```
Initialize student_counter to zero
While student_counter is less than or equal to ten
    Input the next score
    Add the score into the total
EndWhile
Set class_average to total divided by ten
```

Find every variable or loop-control error before proposing a repair.

Discussion 1: What Is Wrong?

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```
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Set class_average to total divided by ten
```

What was this originally?

Discussion 1: What Is Wrong?

The algorithm should read 10 scores and compute their average.

```
Initialize student_counter to zero
While student_counter is less than or equal to ten
    Input the next score
    Add the score into the total
EndWhile
Set class_average to total divided by ten
```

Will this loop end?

Discussion 2: Fill in the Decisions

```
Set passes to 0; set failures to 0
Set student_counter to 1
While student_counter <= 10
    Input score
    -----
    -----
Else
    -----
EndIf
    Set student_counter to student_counter + 1
EndWhile
```

Assume the pass mark is available as pass_mark.

Discussion 3: FizzBuzz

For each integer from 1 through 20:

- print `FizzBuzz` if it is divisible by both 3 and 5;
- otherwise print `Fizz` if divisible by 3;
- otherwise print `Buzz` if divisible by 5;
- otherwise print the number.

Discussion 3: FizzBuzz

For each integer from 1 through 20:

- print `FizzBuzz` if it is divisible by both 3 and 5;
- otherwise print `Fizz` if divisible by 3;
- otherwise print `Buzz` if divisible by 5;
- otherwise print the number.

Hint: Order matters!

Discussion 3: FizzBuzz

For each integer from 1 through 20:

- otherwise print `Fizz` if divisible by 3;
- otherwise print `Buzz` if divisible by 5;
- otherwise print the number.

When given as an interview problem, this step is often left out, you're supposed to ask.

Discussion 4: Predict Before Running (1–9)

Assume the lines are executed **in this order**.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

Discussion 4: Predict Before Running (10–18)

Assume the lines are executed **in this order**.

```
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7 = a
a = d
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```

Discussion 5: Class Percentages

Ask for the numbers of boys and girls, then display each percentage.

Sample interaction

```
Enter the number of boys: 65  
Enter the number of girls: 77  
Boys: 46%    Girls: 54%
```

Design the algorithm and flowchart first.

Discussion 6: Route Planning

Start at the hotel, visit every attraction once, and return to the hotel.

- We want a short route, but do not need the absolute optimum.
- Assume a distance matrix already exists.
- Propose a logical, easy-to-follow heuristic in pseudocode.

Discussion 6: Route Planning

Start at the hotel, visit every attraction once, and return to the hotel.

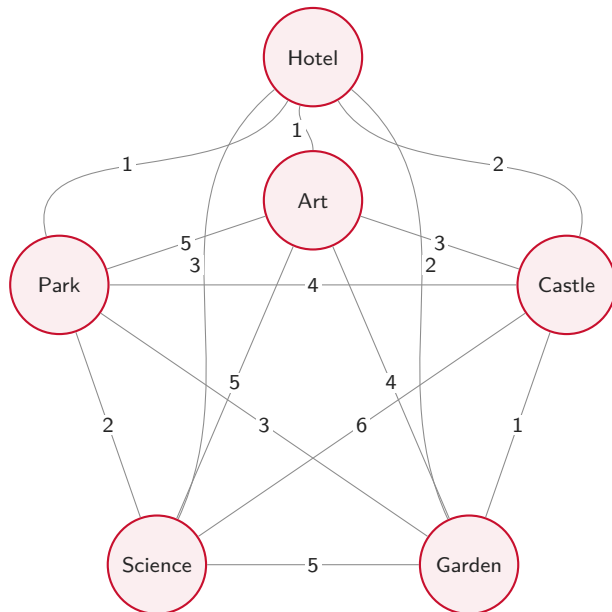
- We want a short route, but do not need the absolute optimum.
- Assume a distance matrix already exists.
- Propose a logical, easy-to-follow heuristic in pseudocode.

Propose a solution to the above problem and describe it using pseudocode. You do NOT need to find the absolute best route, ignore data structure details; assume that the distance matrix has already been created.

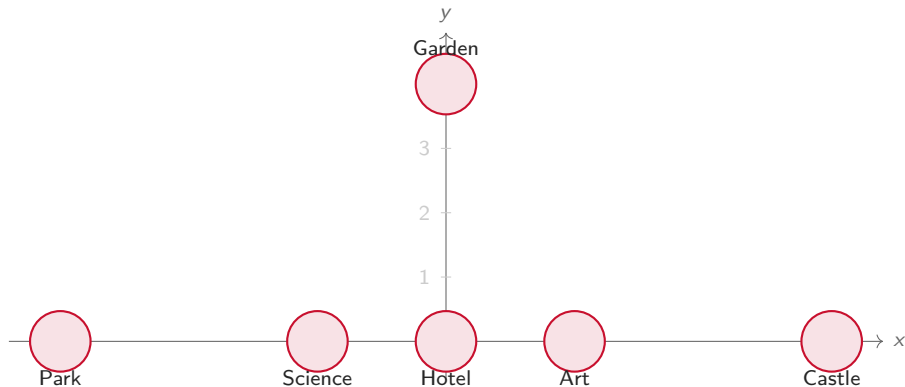
Discussion 6: Distances Between Attractions

	Hotel	Art	Castle	Garden	Science	Park
Hotel	0	1	2	2	3	1
Museum of Art	1	0	3	4	5	5
Historical Castle	2	3	0	1	6	4
Botanical Garden	2	4	1	0	5	3
Science Center	3	5	6	5	0	2
City Park	1	5	4	3	2	0

Discussion 6: The Distance Graph



Discussion 6: A Counterexample in the Plane



Points: Hotel $(0, 0)$, Art $(1, 0)$, Castle $(3, 0)$, Garden $(0, 4)$, Science $(-1, 0)$, Park $(-3, 0)$.

Discussion 6: Counterexample Distance Matrix

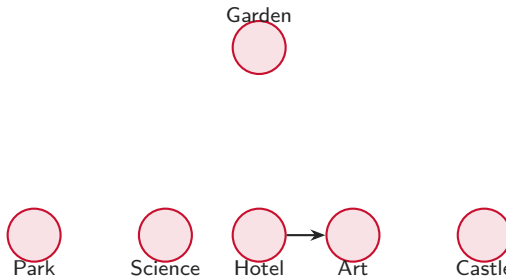
	Hotel	Art	Castle	Garden	Science	Park
Hotel	0.00	1.00	3.00	4.00	1.00	3.00
Art	1.00	0.00	2.00	4.12	2.00	4.00
Castle	3.00	2.00	0.00	5.00	4.00	6.00
Garden	4.00	4.12	5.00	0.00	4.12	5.00
Science	1.00	2.00	4.00	4.12	0.00	2.00
Park	3.00	4.00	6.00	5.00	2.00	0.00

These are Euclidean distances rounded to two decimal places.

Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel $\rightarrow$ Art because both Art and Science are distance 1.00.

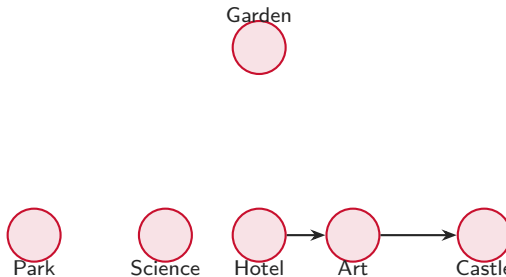


Running total grows as the path commits to each local choice.

Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel $\rightarrow$ Art because both Art and Science are distance 1.00.
- Art $\rightarrow$ Castle because 2.00 is the smallest remaining distance.

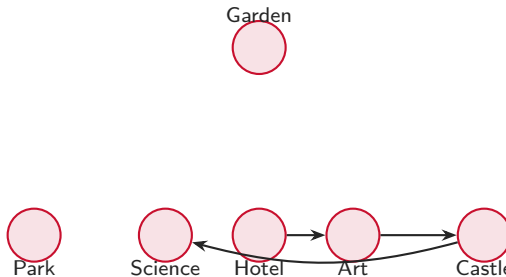


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- Art $\rightarrow$ Castle because 2.00 is the smallest remaining distance.
- Castle $\rightarrow$ Science because 4.00 beats 5.00 and 6.00.

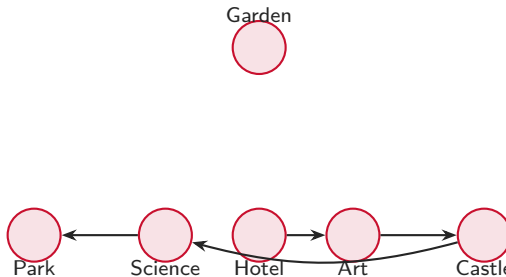


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- Castle $\rightarrow$ Science because 4.00 beats 5.00 and 6.00.
- Science $\rightarrow$ Park because 2.00 is smallest.

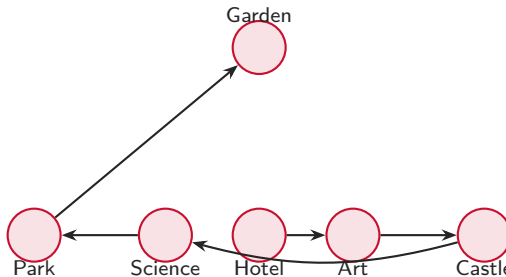


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- Science $\rightarrow$ Park because 2.00 is smallest.
- Park $\rightarrow$ Garden is forced, with distance 5.00.



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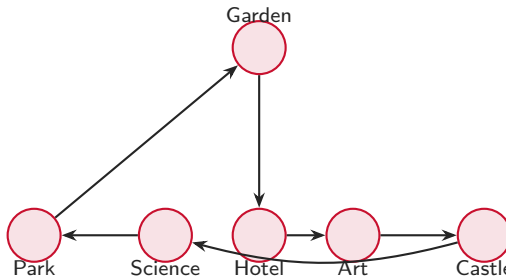
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- Art $\rightarrow$ Castle because 2.00 is the smallest remaining distance.
- Castle $\rightarrow$ Science because 4.00 beats 5.00 and 6.00.
- Science $\rightarrow$ Park because 2.00 is smallest.
- Park $\rightarrow$ Garden is forced, with distance 5.00.
- Garden $\rightarrow$ Hotel closes the tour.

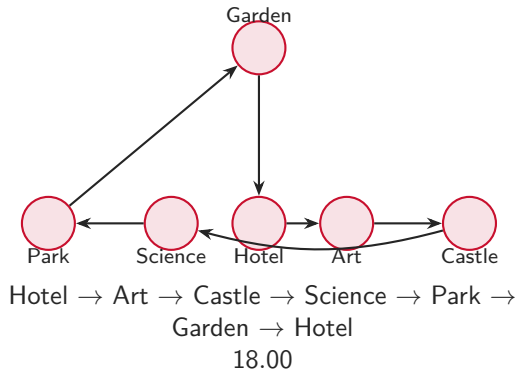
Total greedy length:

$$1.00 + 2.00 + 4.00 + 2.00 + 5.00 + 4.00 = 18.00.$$



Discussion 6: Greedy vs. Better Tour

Greedy choice



Shorter tour

